Local AI on the GPU in C#

A YOLO face model on my laptop GPU: 10 ms a frame. Community YOLO models from Hugging Face, ONNX Runtime and DirectML, no cloud, no Python at runtime. Then from faces to tracks to people.



C# · .NET 10 · YOLO · Hugging Face · ONNX Runtime · DirectML

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Three steps, all on the PC

Pretrained model

an .onnx file from Hugging Face



ONNX Runtime

inside the C# app



GPU · DirectML

the CPU if the GPU can't load it



Face boxes

back on the 4K frame

One frame's journey through the model

1. **Letterbox**
   scale the 4K frame into 640×640; grey bars keep faces' shape

2. **Tensor**
   RGB planes, each value ÷ 255: 1 × 3 × 640 × 640 floats, 4.9 MB, reused

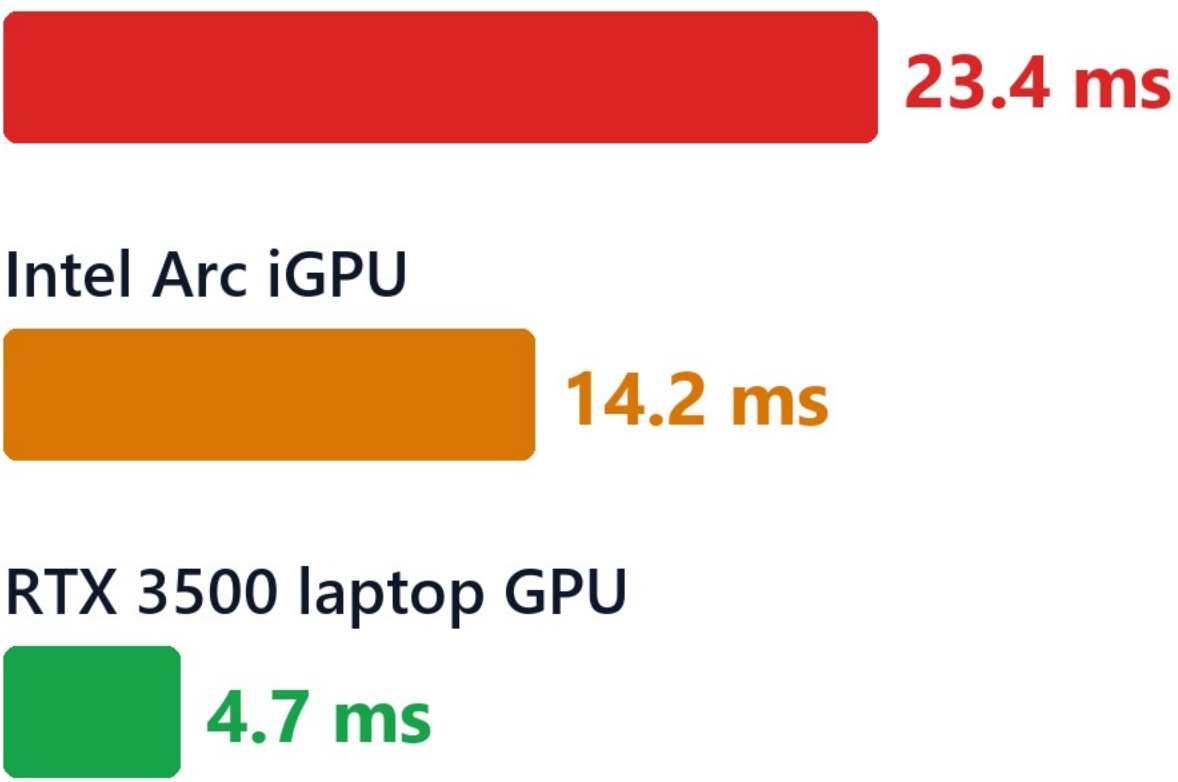
3. **Inference**
   YOLO11n on the GPU returns 8,400 candidate boxes

4. **Filter**
   keep scores over 0.35

5. **Map back, then NMS**
   undo the letterbox onto the 4K frame, then drop overlapping duplicates (IoU 0.4)

CPU vs GPU, measured

YOLO11n, one 1080p frame.



The RTX is 5× faster than the CPU; the integrated GPU 1.65×. At 4K the decoder sets the pace: the 15 s clip scans in 5.2 s on the GPU, 5.9 s on the CPU.

Three models, the same street



UltraFace

4.2 faces per frame

3.1 ms per frame (RTX)



YOLO11n

9.1 faces per frame

5.4 ms per frame (RTX)



YOLO26l

25.0 faces per frame

74.0 ms per frame (RTX)

Averages over 420 frames. No ground truth: more boxes is not the same as more correct.

Check the model's format when it loads

This YOLO26 model was exported NMS-free: its output is the final boxes, [1, 300, 6]. A classic YOLO decoder would read the corners as centres and sizes: wrong boxes, no error.

YoloProfile

needs 8,400 candidate rows at 640×640 ($80^2 + 40^2 + 20^2$); otherwise it throws InvalidModel with a hint: an NMS-free export?

Each family (UltraFace, Yolo, YoloEndToEnd) has a profile that picks its preprocessing and decoder: the Strategy pattern.

The patterns doing the work

Strategy

`IModelFormatProfile`

a profile per model family

Decorator

`NmsDecoder(YoloDecoder)`

NMS wraps a decoder

Composite

`DetectorHost`

several models answer as one

Factory + fallback

`OnnxDetectorFactory`

DirectML first, then the CPU

Trade-offs of local inference

- DirectML is in maintenance mode: Windows ML is one adapter change away.
- Licences: Ultralytics-trained YOLO models are AGPL-3.0 by Ultralytics' terms (the YOLO11n card says Apache-2.0); UltraFace is MIT; the WIDER FACE training data is non-commercial.
- 5 analysed frames a second: the tracker fills the gaps.
- At 4K the decoder sets the pace, not the model.

Detections → tracks → people

Detections

the AI scans 5 frames a second

Tracker

links each face from scan to scan

Tracks

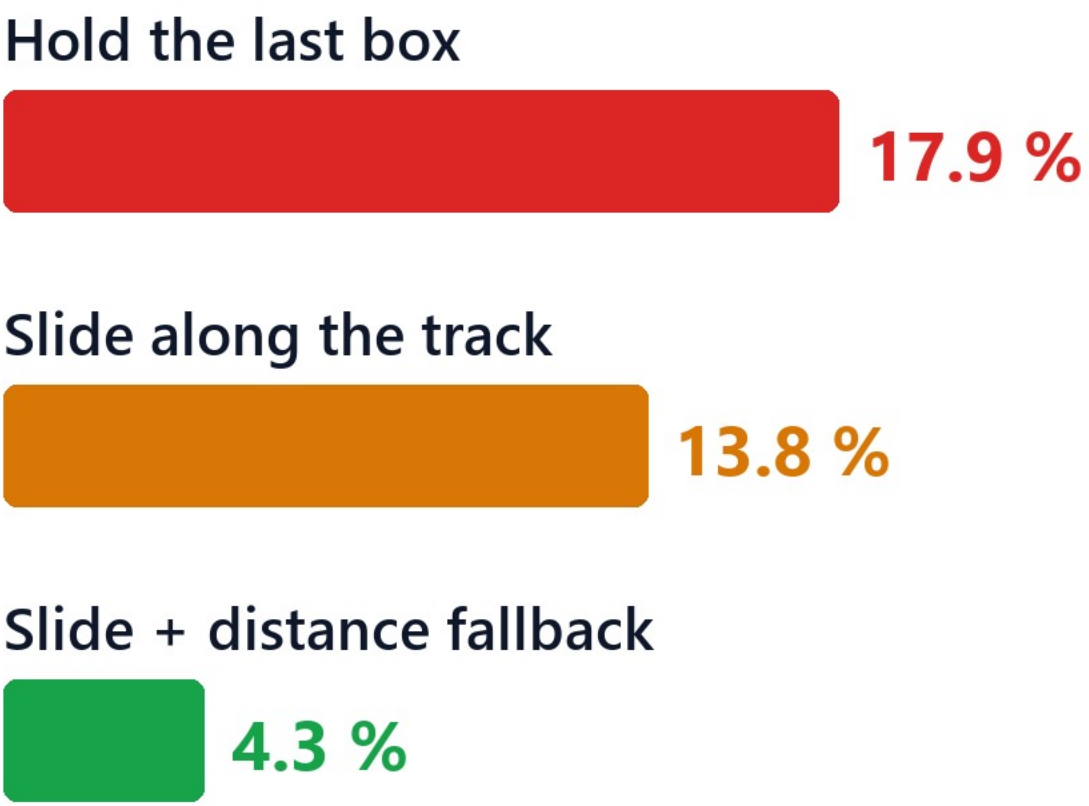
a number that follows the face

People

optional: grouped on the device

Between scans, slide the box

Holding the last box lets a moving face walk out of it. Sliding it towards the same track's next box keeps it on.



Boxes more than 8 px off the face, one minute of 1080p60 footage (my drift benchmark).

Small, fast faces broke tracks

Overlap (IoU ≥ 0.3) fails when a small face moves half its width or more between scans: every scan started a new track.

262 → **80**

tracks in one minute of footage

A distance fallback: a close centre (within 1.5 box sizes) and a similar size keep the track. Overlap matches still win.

Ids that never repeat

A re-scan keeps reviewed tracks. If the tracker restarted at T1, a new face could share an id with an approved one.

T1 T2 T3 T4 approved, kept

High-water mark

the next id is one above the highest ever used; deletes never lower it.

Found by reasoning, pinned by four tests; it never happened in real footage.

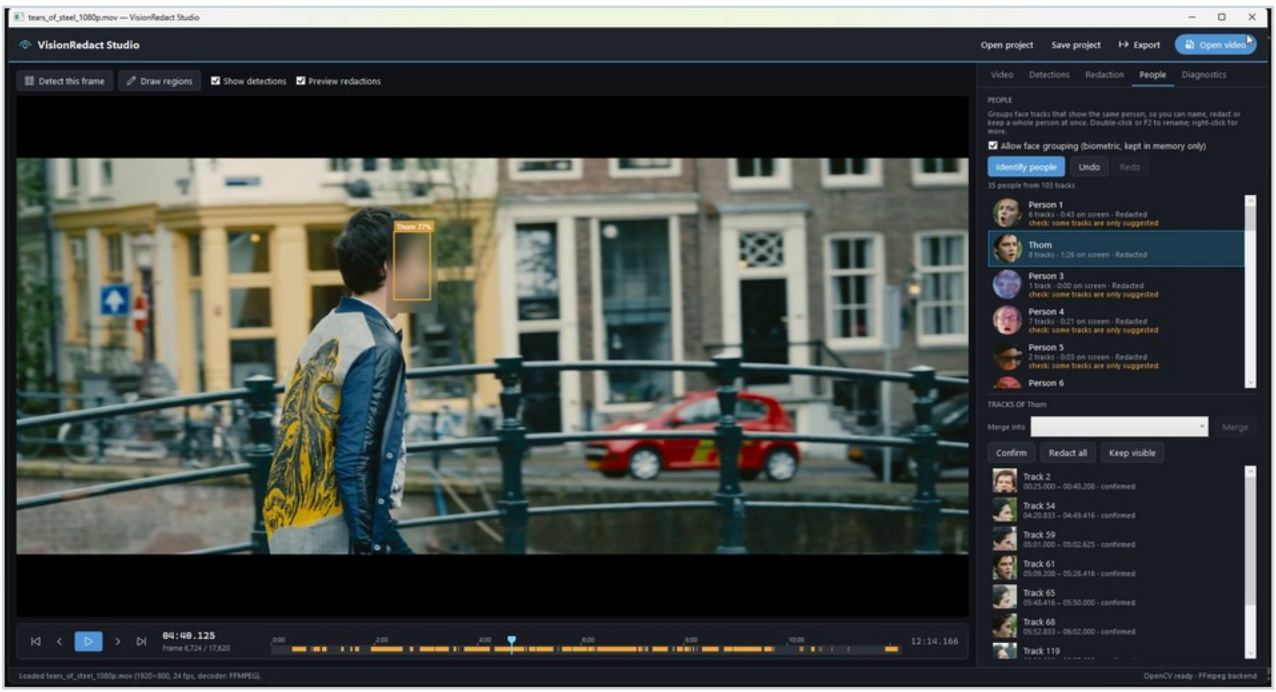
From tracks to people, on the device

1. **Shots**
   3 sharp faces per track
2. **Align**
   YuNet finds 5 landmarks
3. **Embed**
   ArcFace: 512 numbers per face
4. **Group**
   close numbers → a suggested person
5. **Rule**
   the app never groups two tracks from one frame

Opt-in. Embeddings stay in memory; projects never store them.

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Suggestions, confirmations, undo



- Rename, confirm or merge: those choices are never overridden by later suggestions.
- Every edit is a new immutable snapshot, so undo is "go back one".

Trade-offs, and what's next

- No appearance model in the tracker: crossing faces can swap tracks.
- Face grouping is biometric: opt-in, memory only; thumbnails are shown unblurred.
- The ArcFace model is non-commercial; SFace (Apache-2.0 files) is the likely swap, once its training data checks out.
- KCF follows a drawn box forward, 21 ms a frame on the CPU.

Next episode

Fail-safe review and a parallel 4K export pipeline in C# and FFmpeg.

Where it's going: the whole pipeline on the GPU.